Solving Low-Rank Representation Through Deep Newton Networks

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Joint work with H. Chen (SUEP), X. Wang (SUEP), R. Yu (SUEP) and W. Liu (SYSU)

Outline

Introduction

Proposed Methods

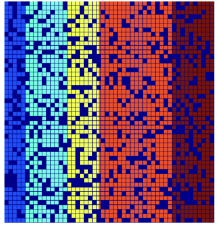
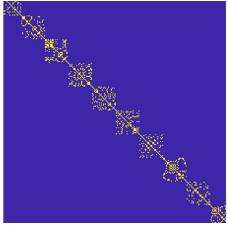
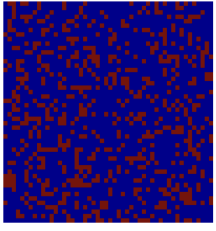
Numerical Experiments

Conclusions and Future Work

LRR

- Liu-Lin-Yan et al, Robust Recovery of Subspace Structures by Low-Rank Representation, IEEE TPAMI, 2013

$$\min_{Z,E} \ \operatorname{rank}(Z) + \lambda\|E\|_0 \quad \text{s.t.} \ \ X = XZ + E \qquad \Rightarrow \qquad \min_{Z,E} \ \|Z\|_* + \lambda\|E\|_1 \quad \text{s.t.} \ \ X = XZ + E$$

- Board applications: clustering, classification, denoising, detection, fusion
- Related to low-rank regularization, low-rank approximation, low-rank decomposition

DU

- Gregor-LeCun, Learning Fast Approximations of Sparse Coding, ICML, 2010

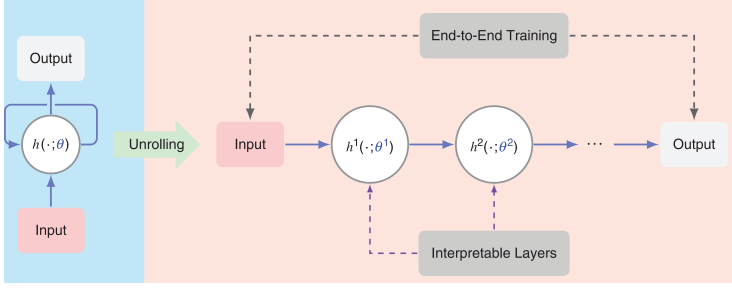
$$\min_{x\in\mathbb{R}^n} \frac{1}{2}\|Ax-b\|^2+\lambda\|x\|_1$$

$$\Downarrow$$

$$x^{k+1}=\texttt{Prox}_{\lambda\|\cdot\|_1}(x^k-\rho A^\top(Ax^k-b))$$

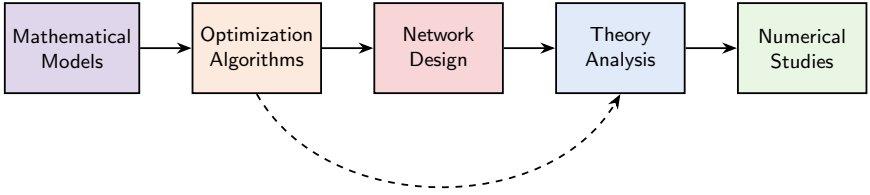
$$\Downarrow$$

$$x^{k+1}=h_{\theta^k}(x^k-\rho A^\top(Ax^k-b))$$



Motivation

- Chen-Liu-Wang et al, Theoretical Linear Convergence of Unfolded ISTA and Its Practical Weights and Thresholds, NeurIPS, 2018
- Yang-Sun-Li et al, ADMM-CSNet: A Deep Learning Approach for Image Compressive Sensing, IEEE TPAMI, 2020
- Chen-Chen-Chen et al, Learning to Optimize: A Primer and A Benchmark, JMLR, 2022
- He-Xiu, Newton Deep Unfolding for Compressed Sensing, arXiv:2609.14391
- Xiu-He-Qu-Sun, A Survey of Deep Unfolding Methods for Image Compressive Sensing (in Chinese), ORT, 2026



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Model

- Consider the following LRR model

$$\begin{aligned} \min_{Z,E} \quad & \|Z\|_* + \lambda \|E\|_1 \\ \text{s.t.} \quad & X = XZ + E \end{aligned}$$

- Augmented Lagrangian method (ALM)

$$\begin{aligned} \min_{J,E,Z} \quad & \|J\|_* + \lambda \|E\|_1 \\ \text{s.t.} \quad & X = XZ + E, \ Z = J \end{aligned}$$

which derives

$$\begin{aligned} \mathcal{L}_\beta(J, E, Z, Y_1, Y_2) = & \|J\|_* + \lambda \|E\|_1 + \langle Y_1, Z - J \rangle \\ & + \frac{\beta}{2} \|Z - J\|_F^2 + \langle Y_2, X - XZ - E \rangle + \frac{\beta}{2} \|X - XZ - E\|_F^2 \end{aligned}$$

Algorithm

- Update variables according to

$$\begin{cases} (J, E, Z) \leftarrow \arg\min_{J,E,Z} \ \mathcal{L}_\beta(J, E, Z, Y_1, Y_2) \\ Y_1 \leftarrow Y_1 + \beta(Z - J) \\ Y_2 \leftarrow Y_2 + \beta(X - XZ - E) \end{cases}$$

- First-order optimality conditions

$$\begin{cases} 0 \in \partial\|J\|_* - Y_1 + \beta(J - Z) \\ 0 \in \lambda\,\partial\|E\|_1 - Y_2 - \beta(X - XZ - E) \\ 0 = Y_1 + \beta(Z - J) - X^\top Y_2 - \beta X^\top (X - XZ - E) \end{cases}$$

which derives

$$J = \mathtt{Prox}_{1/\beta, \|\cdot\|_*}\left(Z + Y_1/\beta\right), \quad E = \mathtt{Prox}_{\lambda/\beta, \|\cdot\|_1}\left(X - XZ + Y_2/\beta\right)$$

SSN

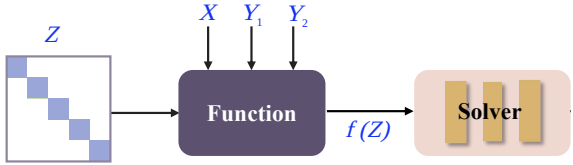
- Denote the nonlinear equation

$$
\begin{aligned}
f(Z) = Y_1 + \beta\left(Z - \mathtt{Prox}_{1/\beta,\|\cdot\|_*}(Z + Y_1/\beta)\right) - X^\top Y_2 \\
- \beta X^\top\left(X - XZ - \mathtt{Prox}_{\lambda/\beta,\|\cdot\|_1}(X - XZ + Y_2/\beta)\right)
\end{aligned}
$$

$$
\Downarrow
$$

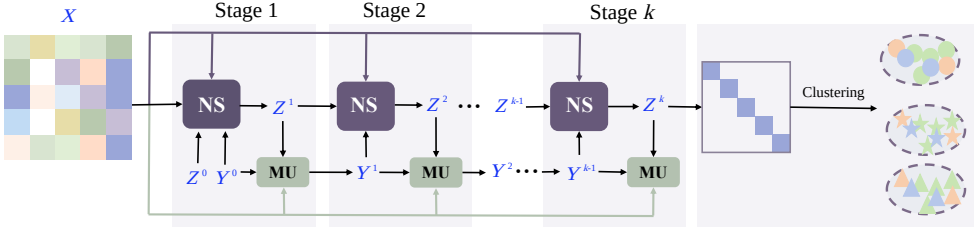
$$
\begin{cases}
g(Z)\Delta Z = -f(Z) \\
Z \leftarrow Z + \eta \Delta Z
\end{cases}
$$

- Newton solver (NS)



LRRDNN

- Overall architecture



- Training loss

$$\ell_{\text{total}} = \ell_{\text{self}} + \sigma_1 \ell_{\text{fun}} + \sigma_2 \ell_{\text{reg}}$$

$$= \left(\|X - XZ^K - E^K\|_F^2 + \gamma \|E^K\|_1 \right) + \sigma_1 \left(\frac{1}{N} \sum_{l=1}^{N} \|f(Z^K)\|_1 \right) + \sigma_2 \text{tr}(L_{Z^K})$$

LRRDNN+

- How to construct regularization terms?

$$\begin{aligned} \min_{Z,E} \quad & \|Z\|_* + \lambda \|E\|_1 \\ \text{s.t.} \quad & X = XZ + E \end{aligned} \quad \Rightarrow \quad \begin{aligned} \min_{Z,E} \quad & \mathcal{R}(Z) + \lambda \mathcal{S}(E) \\ \text{s.t.} \quad & X = XZ + E \end{aligned}$$

- Similar ALM and first-order optimality conditions

$$\begin{cases} 0 \in \partial \mathcal{R}(J) - Y_1 + \beta(J - Z) \\ 0 \in \lambda \, \partial \mathcal{S}(E) - Y_2 - \beta(X - XZ - E) \\ 0 = Y_1 + \beta(Z - J) - X^\top Y_2 - \beta X^\top (X - XZ - E) \end{cases}$$

which derives

$$J = \texttt{Prox}_{1/\beta, \mathcal{R}} \left(Z + Y_1/\beta \right), \quad E = \texttt{Prox}_{\lambda/\beta, \mathcal{S}} \left(X - XZ + Y_2/\beta \right)$$

LRRDNN+

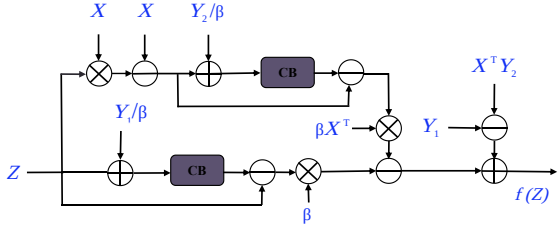
- Denote the nonlinear equation

$$f(Z) = Y_1 + \beta\left(Z - \texttt{Prox}_{1/\beta,\mathcal{R}}(Z + Y_1/\beta)\right) - X^\top Y_2 - \beta X^\top\left(X - XZ - \texttt{Prox}_{\lambda/\beta,\mathcal{S}}(X - XZ + Y_2/\beta)\right)$$

$$\Downarrow$$

$$\begin{cases} g(Z)\Delta Z = -f(Z) \\ Z \leftarrow Z + \eta\Delta Z \end{cases}$$

- Convolutional block (CB)



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Experiments

- Compared methods
  - `TLRR`: Zhou-Lu-Feng et al, IEEE TPAMI, 2021
  - `ETLRR`: Du-Liu-Shan et al, KBS, 2022
  - `IDR`: Wei-Liu-Zhou et al, IEEE TKDE, 2023
  - `TS1-LLRR`: Hu-Qu-Liu et al, KBS, 2024
  - `JSSC`: Zhu-Xiu-Liu et al, ESWA, 2025
  - `TFLLRR`: Liu-Hu-Wang, IEEE TSIPN, 2025
  - `BTLRR`: Ding-Zhao-Yang et al, SIIMS, 2025
  - `ISC`: Zhang-Zhou-Yao et al, IEEE TPAMI, 2026
- Implementation details
  - Hyperparameters: $\sigma_1 = 0.01$, $\sigma_2 = 1$, K=6
  - Learnable parameters: λ, β, $\mathcal{R}(J)$, $\mathcal{S}(E)$,
  - Clustering: k-means
  - Metrics: ACC ($\uparrow$)
- https://xianchaoxiu.github.io

Results

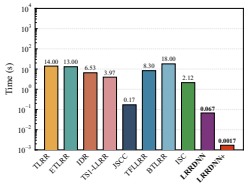
- Clustering comparisons

Datasets	TLRR	ETLRR	IDR	TS1-LLRR	JSCC	TFLLRR	BTLRR	ISC	LRRDNN	LRRDNN+
ORL	59.03%	58.24%	73.32%	73.46%	74.53%	68.70%	53.53%	74.75%	75.25%	**90.25%**
PalmData25	77.94%	81.89%	95.27%	91.25%	93.80%	93.55%	92.52%	93.55%	**97.45%**	95.95%
FRDUE	74.23%	84.18%	95.31%	95.31%	91.35%	96.71%	94.66%	96.65%	**97.24%**	96.02%
USPS	40.97%	62.69%	79.80%	79.80%	69.62%	74.20%	55.59%	65.30%	80.50%	**99.40%**
LUNG	64.33%	42.91%	86.21%	75.81%	85.52%	63.04%	66.01%	74.88%	**87.69%**	79.31%
EYaleB	80.68%	76.99%	81.28%	73.86%	85.96%	71.27%	79.94%	75.59%	83.21%	**99.09%**
MNIST	33.44%	30.37%	59.59%	58.61%	46.60%	48.27%	37.65%	32.09%	54.93%	**93.43%**
AR	36.51%	12.79%	72.64%	76.46%	78.52%	71.67%	61.40%	54.42%	73.14%	**89.95%**
Average	58.39%	56.26%	80.43%	78.07%	78.24%	73.43%	67.66%	70.90%	81.18%	**92.93%**

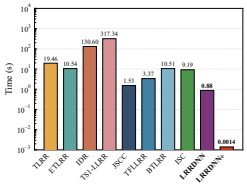
Models	SSN	ORL	PalmData25	FRDUE	USPS	LUNG	EYaleB	MNIST	AR	Average
LRRDNN	×	73.50%	91.75%	92.96%	76.10%	86.70%	79.79%	52.17%	72.61%	78.20%
	✓	**75.25%**	**97.45%**	**97.24%**	**80.50%**	**87.69%**	**83.21%**	**54.93%**	**73.14%**	**81.18%**
LRRDNN+	×	90.50%	94.00%	92.60%	97.10%	91.63%	98.81%	89.65%	82.69%	92.12%
	✓	**98.75%**	**99.35%**	**98.72%**	**99.40%**	**94.09%**	**99.09%**	**93.43%**	**89.95%**	**96.60%**

Efficiency

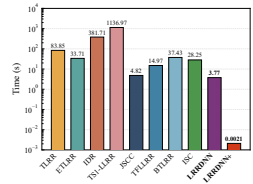
- Runtime in a logarithmic scale



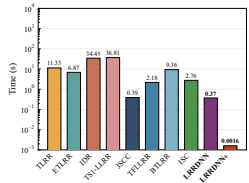
(a) ORL



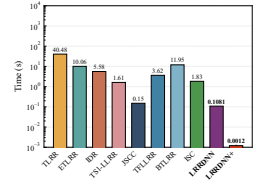
(b) PalmData25



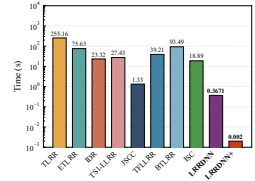
(c) FRDUE



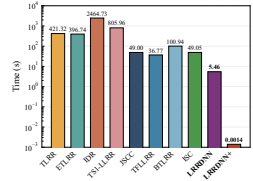
(d) USPS



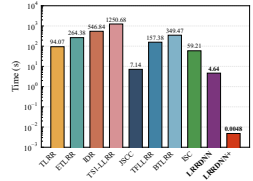
(e) LUNG



(f) EYaleB



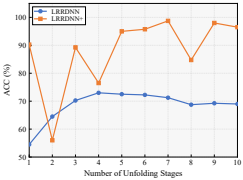
(g) MNIST



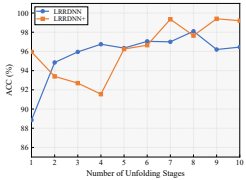
(h) AR

Analysis

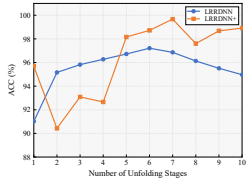
- Effect of the number of unrolling stages K



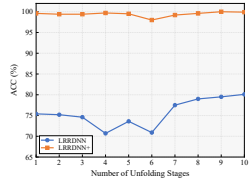
(i) ORL



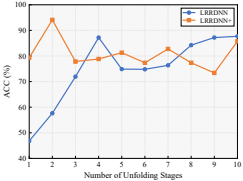
(j) PalmData25



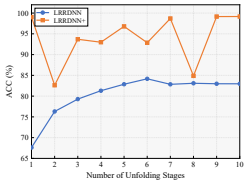
(k) FRDUE



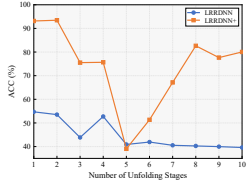
(l) USPS



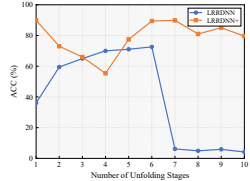
(m) LUNG



(n) EYaleB



(o) MNIST



(p) AR

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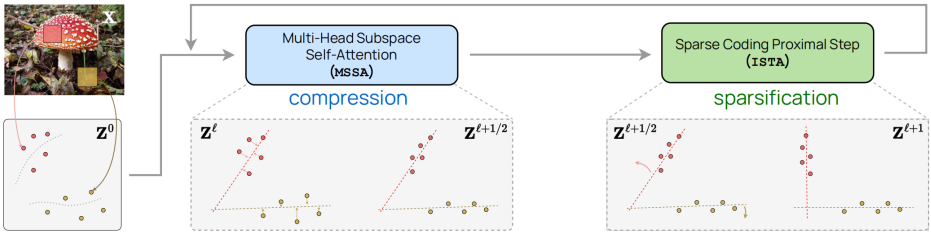
Conclusions

- Chen-Wang-Xiu-Liu, Rethinking Low-Rank Representation with Deep Semi-Smooth Newton, PRCV, 2026
- Chen-Yu-Wang-Xiu-Liu, Solving Low-Rank Representation Through Deep Newton Networks, working in process
- Chen-Wang-Xiu-Liu, LLM-IMVC: Large Language Model-Guided Deep Unrolling for Incomplete Multi-View Clustering, working in process

Methods	HandWrite	BDGP	MSRC	COIL20	Caltech5V	USPS	Average
LMVSC (AAAI'20)	67.1%	55.6%	35.2%	69.2%	46.1%	56.0%	54.9%
SFMC (TPAMI'22)	75.7%	37.8%	60.0%	74.8%	46.8%	98.9%	65.7%
FPMVS-CAG (TIP'22)	82.3%	55.6%	80.5%	60.2%	80.9%	78.3%	73.0%
FDAGF (AAAI'23)	82.3%	52.5%	73.9%	74.0%	81.3%	56.1%	70.0%
RCAGL (TKDE'24)	79.4%	51.8%	77.6%	65.1%	83.6%	60.9%	69.7%
ALPC (AAAI'25)	63.2%	92.2%	64.3%	76.4%	46.1%	50.2%	65.4%
TLRLF4 (TPAMI'25)	76.4%	92.7%	—	76.3%	75.5%	99.2%	84.0%
LLM-IMVC	**99.9%**	**100.0%**	**88.1%**	**77.2%**	**100.0%**	**99.9%**	**94.2%**

Future Work

- Yang-Huang-Wipf, Transformers from An Optimization Perspective, NeurIPS, 2022
- Yu-Buchanan-Pai et al, White-box Transformers via Sparse Rate Reduction, NeurIPS, 2023
- Porras-Hadou-Ribeiro, A Constrained Optimization Perspective of Unrolled Transformers, ICML, 2026
- Tai-Liu-Li et al, A Mathematical Explanation of Transformers, SIIMS, 2026



Thank you for your attention!

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